7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Unit 7, Lesson 4: Trigonometric Ratios on the Coordinate Plane

Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

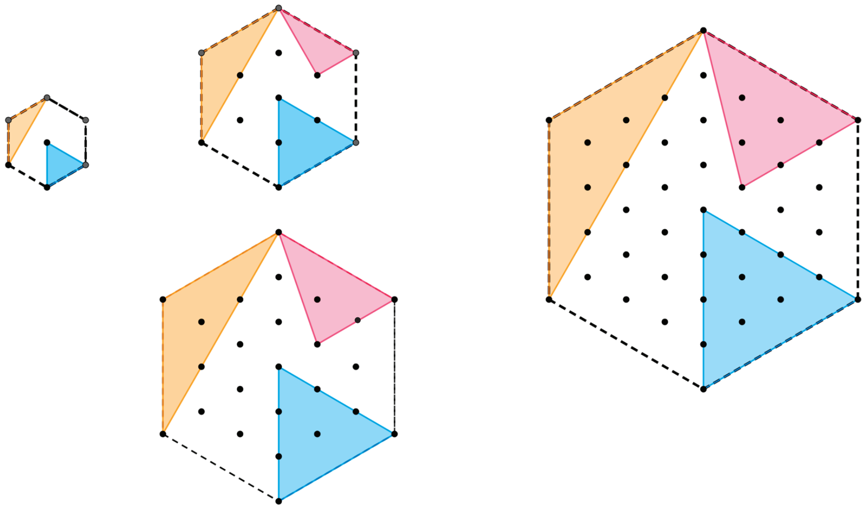
MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Target

- ☐ I can determine trigonometric ratios using the coordinate plane.

7.4.1 Warm-Up: Patterns

Four hexagons are shown.

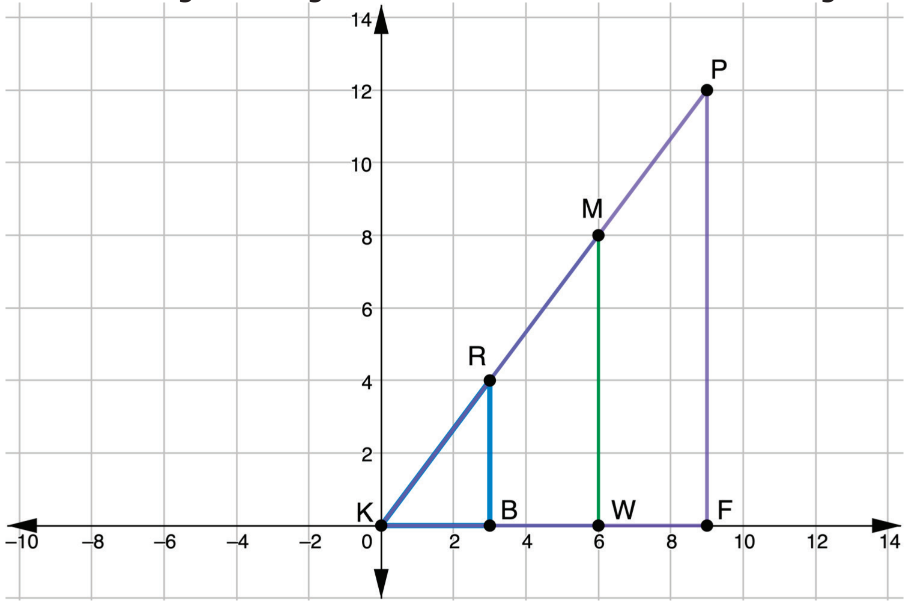


1. Describe the pattern of each triangle.

Triangle	Pattern Description
Yellow	
Red	
Blue	

7.4.2 Collaboration: Defining Trigonometric Ratios

1. Three right triangles are shown on the coordinate grid.



a. Discuss with your partner how to justify that ΔKBR, ΔKWM, and ΔKFP are similar.
Summarize your discussion.

b. Complete the table.

Segment	Length	Segment	Length
$\overline{KB}$		$\overline{WM}$	
$\overline{BR}$		$\overline{KF}$	
$\overline{KW}$		$\overline{FP}$	

c. Use the lengths from the table to determine each. Show your work.

Segment	Length
$\overline{KR}$	
$\overline{KM}$	
$\overline{KP}$	

d. Using $\angle K$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
ΔRKB			
ΔMKW			
ΔPKF			

2. ΔHWB is graphed on the coordinate grid, where $m\angle WBH = 90°$. The coordinates of the vertices are $H\ (-4, 5)$, $W\ (5, -3)$, and $B\ (-4, -3)$.

   a. Discuss with your partner how to use the coordinates to determine the trigonometric ratios. Summarize your discussion.

   b. Complete the table. Show your work.

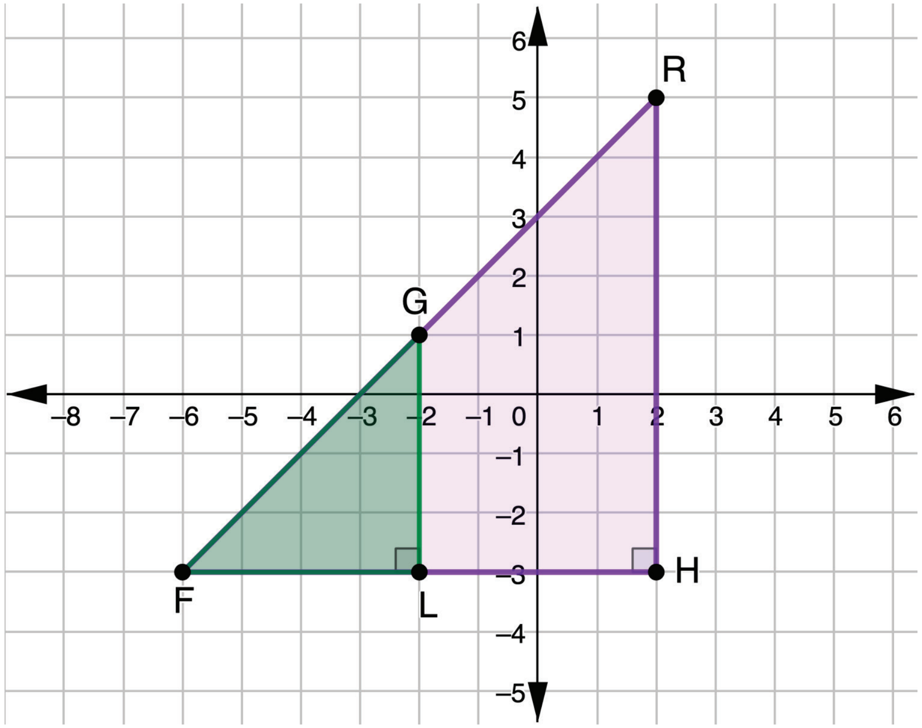
Segment	Length
$\overline{WB}$	
$\overline{BH}$	
$\overline{HW}$	

   c. Using $\angle WHB$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

7.4.3 Guided Instruction: Using Trigonometric Ratios

1. ΔFLG and ΔFHR shown on the coordinate grid are isosceles right triangles.



a. Determine the measure of each angle.

Angle	Measure
$\angle LFG$	
$\angle FGL$	
$\angle FRH$	

b. Using $\angle F$ as the reference angle, find the three ratios for each triangle.

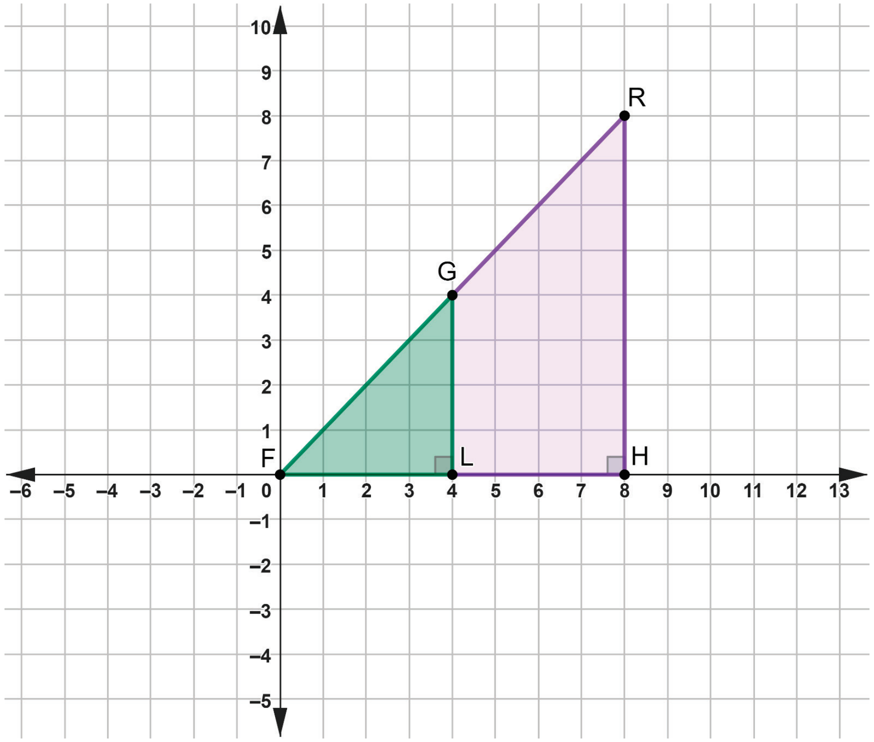
Triangle	Cosine	Sine	Tangent
ΔFLG			
ΔFHR			

c. Use your calculator to find each. Round to the nearest thousandth, if necessary.

Angle	Cosine	Sine	Tangent
$45°$			

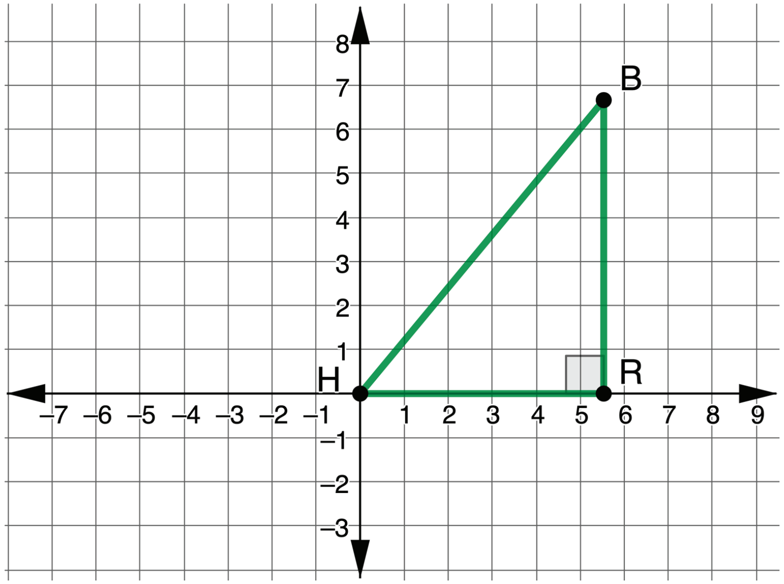
d. How are the ratios you found using $\angle F$ as the reference angle related to the values you found for the ratios of a $45°$ angle?

e. ΔFLG and ΔFHR were moved on the coordinate grid so that F is at the origin.



Work with your partner to determine whether or not the values of the trigonometric ratios will change because the triangle moved. Summarize your findings.

2. ΔHBR is shown on the coordinate grid, where $m\angle HRB = 90°$, $m\angle RHB = 50°$, $HR = 5.567$, and $HB = 8.66$.



a. Explain why the ratio $\cos(50°) \approx \frac{5.567}{8.66}$ is true.

b. Find each value using a calculator. Round to the nearest thousandth.

$\cos(50°) =$ $\qquad\qquad$ $\frac{5.567}{8.66} =$

c. Discuss with your partner why the approximation symbol was used in $\cos(50°) \approx \frac{5.567}{8.66}$. Summarize your thoughts below.

d. Work with your partner to find $\sin(H)$. Round to the nearest thousandth.

e. Use your calculator to find $\sin(50°)$.

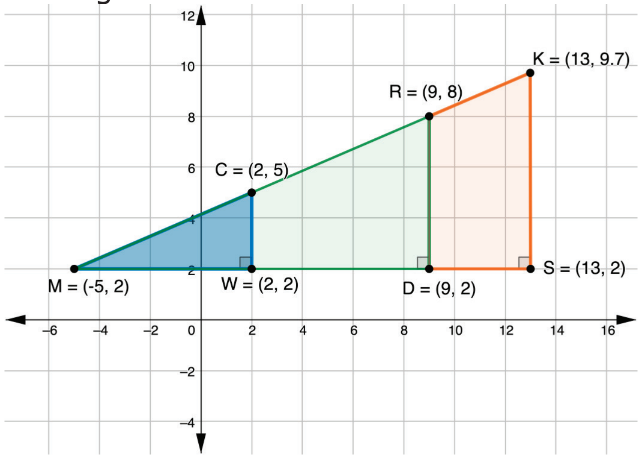
f. Explain why $\sin(50°) \approx \sin(H)$.

g. Work with your partner to write the coordinates of each vertex.

Vertex	Coordinates
R	
B	

7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

1. ΔCMW, ΔRMD, and ΔKMS are shown on the coordinate grid.



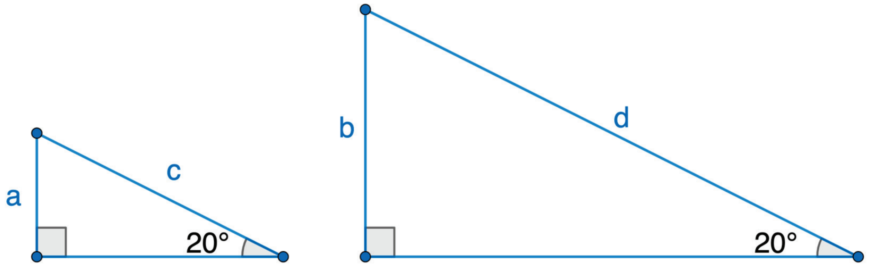
Using $\angle M$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

Triangle	Cosine	Sine	Tangent
ΔCMW			
ΔRMD			
ΔKMS			

2. Yasmeen said she only found the ratios for ΔCMW then applied them to the other triangles. Explain using mathematics why she can do that.

7.4.5 Wrap-Up: Ticket Out the Door

1. Two triangles are shown.



   a. Describe how the ratios $\frac{a}{c}$ and $\frac{b}{d}$ are related.

   b. What mathematical relationship(s) did you use to complete your description?

Unit 7, Lesson 5: Using Trigonometric Ratios

Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

Learning Targets

- ☐ I can use trigonometric ratios to find the side of a right triangle.
- ☐ I can use trigonometric ratios to find an angle of a right triangle.